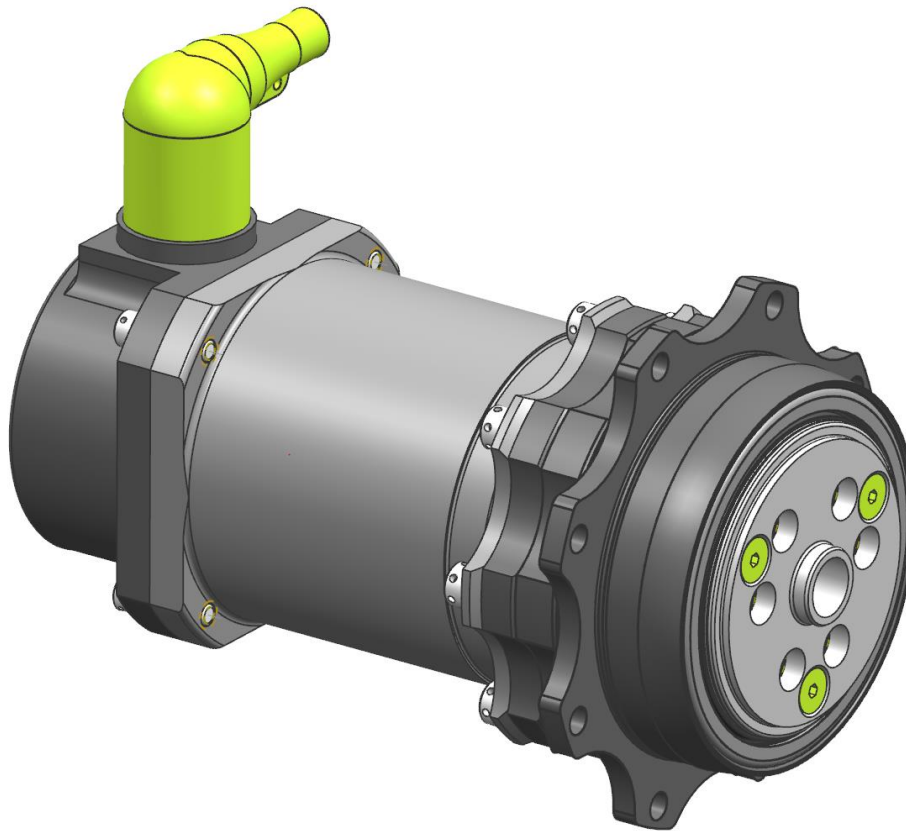
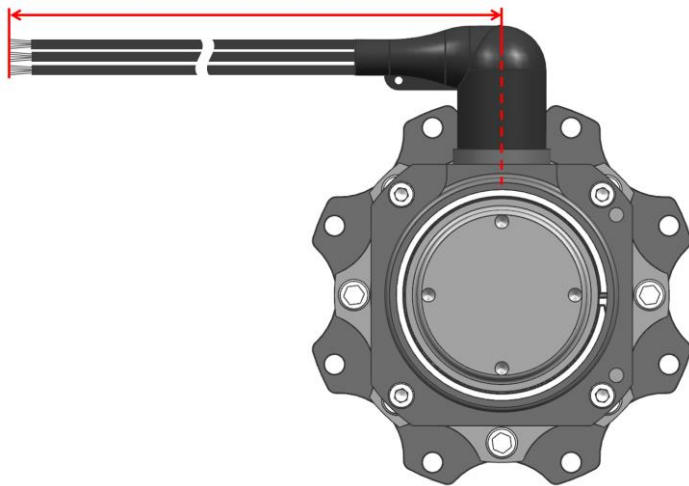


DATASHEET TPM Flight Actuator



Parameter		Unit	Value
Performance			
	Output torque, cont.	Nm	24 ± 10%
	Output torque, cont. stall torque	Nm	26 ± 10%
	Output torque, maximum	Nm	62
	Output speed, max (at 19.2V)	rpm	40
	The Unit is designed for an endurance life of 20000h with @ 24Nm		
Mechanical			
	Size		
	Length	mm	137
	Width	mm	86
	Height	mm	100
	Weight (with 0.92m wiring and I/F connectors)	kg	max. 2,2
Mechanical Interface			
	To structure	8 mounting holes Ø 4,5mm	
	To load	Output shaft centering diameter Ø12,5js7 6x Helicoil 0,25-28 UNF 3B	

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Parameter		Unit	Value
Electrical Interface			
	Cables, length (no connectors attached, see picture below) 	mm	min. 920
	Sine and Cosine output voltage (motor position resolvers)	V _{RMS}	4.5 .. 6.5
	Bus voltage	V _{DC}	19,2 .. 30
	Current (RMS), max.	A	11,92
	Current (RMS), cont.	A	4,91 ± 10%
Motor			
	Topology	3 phase brushless servo	
	Redundancy	winding (2)	
		position sensor (2)	
		temperature sensor (4) (2 per winding)	
	Position sensor	Brushless, single speed resolver (2)	
Temperature sensor	Temperature sensors with negative temperature Coefficient (4)		
Gearbox			
	Topology	Two stage planetary	
	Ratio		100

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TPM Flight Actuator

Classes of environment acc. RTCA DO-160G																			
Item	Class																		
Temperature and Altitude																			
Ground survival low	Sect. 4.5.1, Cat. B4, -55°C																		
Short-Time Operating Low	Sect. 4.5.1, Cat. B4, -54°C																		
Operating Low	Sect. 4.5.2, Cat. B4, -54°C (if powered at -40°C)																		
Ground survival High	Sect. 4.5.3, Cat. B4, +85°C																		
Short-Time Operating High	Sect. 4.5.3, Cat. B4, +85°C																		
Operating High	Sect. 4.5.4, Cat. B4, +85°C																		
Altitude, operational	Sect. 4.6.1, Cat. B4, 20,000ft																		
Altitude, non-operational	Sect. 4.6.1, Cat. B4, 50,000ft (change rate: 33ft/s)																		
Temperature Variation	Sect. 5, Cat. S2, Change rate >139K/min (transfer between chambers in less than 1min)																		
Humidity																			
	Sect. 6, Cat.B, +30..+60°C at constant RH of 95%																		
Shock and Crash Safety																			
Operational Shock	Sect. 7.2, Cat.B, 20g/11ms each axis																		
Crash Safety	Sect. 7.3.1, Test Proc. 1 (Impulse), 40g/11ms each axis Sect. 7.3.3, Test Proc. 2 (Sustained), - operational: all axis 6g except “aircraft up”: 7g																		
Vibration																			
Sine on Random	Sect. 8, Cat. R Random Vibration Profile: 10Hz: 0.002 g²/Hz 10..100Hz: +3.01db/Oct. 100..500Hz: 0.02 g²/Hz > 500Hz: 0 g²/Hz Sine on Random: f₀ = 3.75Hz, A₀ = 0,25g <table><tr><th colspan="2">Freq.</th><th>Amplitude</th></tr><tr><td>f₁</td><td>f₀</td><td>A₀</td></tr><tr><td>f₂</td><td>3 · f₀</td><td>3 · A₀</td></tr><tr><td>f₃</td><td>6 · f₀</td><td>5 · A₀</td></tr><tr><td>f₄</td><td>9 · f₀</td><td>3 · A₀</td></tr><tr><td>f₅</td><td>12 · f₀</td><td>3 · A₀</td></tr></table>	Freq.		Amplitude	f₁	f₀	A₀	f₂	3 · f₀	3 · A₀	f₃	6 · f₀	5 · A₀	f₄	9 · f₀	3 · A₀	f₅	12 · f₀	3 · A₀
Freq.		Amplitude																	
f₁	f₀	A₀																	
f₂	3 · f₀	3 · A₀																	
f₃	6 · f₀	5 · A₀																	
f₄	9 · f₀	3 · A₀																	
f₅	12 · f₀	3 · A₀																	
Explosion Proofness																			
	Sect. 9, Cat. E, n-Hexane																		
Waterproofness																			
	Sect. 10, Cat. W																		
Fluid Susceptibility																			
	Sect. 11, Cat. F Fluids: MIL-PRF-83282, MIL-PRF-5606, Water, Cleaning detergents, CO₂ foam fire fighting material and residue, Fire extinguishing foam concentrate, Alcohol, Isopropyl (TT-I-735), Compound, Cleaning Brulin Safety Solvent #512, MIL-G-81322, MIL-L- 85734, MIL-PRF-23699, MIL-PRF-7808																		

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<i>Sand and Dust</i>	
Dust	Sect. 12, Cat. S airvelocity: 1.5 to 2.5 m/s, Temperature: 6h inop at T _{Ambient} , 6h operative at +71°C
Sand and Dust	Sect. 12, Cat. S airvelocity: 1.5 m/s, Temperature: 6h inop at T _{Ambient} , 6h inop at +71°C
<i>Fungus Resistance</i>	
	Sect. 13, Cat. F
<i>Salt Fog</i>	
	Sect. 14, Cat. S, 24h with four cycles wet/dry
<i>Electro Static Discharge</i>	
	Sect. 25, Cat. A, Testlevel: 25kV
EMC/EMI	i.a.w. RTCA DO-160G, sect.18 to 22 as stated

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